DATABASE DESIGN DOCUMENTS

(DDD)

TOPIC: WAREHOUSE & SUPPLY CHAIN

I/ Use Data Definition Language (DDL) Scripts

```
CREATE DATABASE IF NOT EXISTS lab1;

USE lab1;


CREATE TABLE Warehouse (

    WarehouseID INT NOT NULL AUTO_INCREMENT COMMENT 'Unique identifier of a warehouse',

    WarehouseName VARCHAR(100) NOT NULL COMMENT 'Warehouse name',

    Address VARCHAR(200) NOT NULL COMMENT 'Physical address',

    City VARCHAR(100) NOT NULL COMMENT 'City where the warehouse is located',

    Country VARCHAR(100) NOT NULL COMMENT 'Country where the warehouse is located',

    CapacityM3 DECIMAL(12,2) NOT NULL CHECK (CapacityM3 >= 0) COMMENT 'Maximum storage capacity in cubic meters',


    CONSTRAINT pk_warehouse_id PRIMARY KEY (WarehouseID)

);

CREATE INDEX idx_warehouse_name ON Warehouse(WarehouseName) COMMENT 'Supports searching warehouses by name';


CREATE TABLE Product (

    ProductID   INT NOT NULL AUTO_INCREMENT COMMENT 'Unique identifier of a product',

    ProductName VARCHAR(150) NOT NULL COMMENT 'Product name',

    Description TEXT NULL COMMENT 'Additional product details',

    Unit        VARCHAR(20) NOT NULL COMMENT 'Base unit used to measure quantity',
```

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    ProductType VARCHAR(50) COMMENT 'Product category; allowed values are not finalized in Phase 1',


    CONSTRAINT pk_product_id PRIMARY KEY (ProductID)

);

CREATE INDEX idx_product_name ON Product(ProductName) COMMENT 'Supports product-name lookups';


CREATE TABLE Supplier (

    SupplierID   INT NOT NULL AUTO_INCREMENT COMMENT 'Unique identifier of a supplier',

    SupplierName VARCHAR(150) NOT NULL COMMENT 'Supplier business name',

    ContactName  VARCHAR(100) COMMENT 'Main contact person',

    Phone        VARCHAR(30) COMMENT 'Supplier contact telephone number',

    Email        VARCHAR(150) COMMENT 'Supplier contact email; not unique unless the business requires it',

    Address      VARCHAR(200) COMMENT 'Supplier business address',


    CONSTRAINT pk_supplier_id PRIMARY KEY (SupplierID)

);

CREATE INDEX idx_supplier_name ON Supplier(SupplierName) COMMENT 'Supports supplier-name lookups';


CREATE TABLE Inventory (

    InventoryID INT NOT NULL AUTO_INCREMENT COMMENT 'Unique inventory record identifier',

    WarehouseID INT NOT NULL COMMENT 'Warehouse holding the product',

    ProductID   INT NOT NULL COMMENT 'Product held at the warehouse',

    Quantity    DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (Quantity >= 0) COMMENT 'Current stock balance',

    ReorderLevel DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (ReorderLevel >= 0) COMMENT 'Threshold below which a restock alert is opened',
```

```
    LastUpdated DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON
UPDATE CURRENT_TIMESTAMP COMMENT 'Last inventory record update time',


    CONSTRAINT pk_inventory_id PRIMARY KEY (InventoryID),

    CONSTRAINT uq_inventory_warehouseid_productid UNIQUE (WarehouseID,
ProductID),

    CONSTRAINT fk_inventory_warehouse_id FOREIGN KEY (WarehouseID)
REFERENCES Warehouse(WarehouseID),

    CONSTRAINT fk_inventory_product_id FOREIGN KEY (ProductID)
REFERENCES Product(ProductID)

);


CREATE TABLE SupplierProduct (

    SupplierID   INT NOT NULL COMMENT 'Supplier offering the product',

    ProductID    INT NOT NULL COMMENT 'Product offered by the supplier',

    UnitPrice    DECIMAL(12,2) NOT NULL CHECK (UnitPrice >= 0) COMMENT
'Current price offered by this supplier',

    LeadTimeDays INT NOT NULL DEFAULT 0 CHECK (LeadTimeDays >= 0)
COMMENT 'Expected supplier delivery lead time in days',


    CONSTRAINT pk_supplierproduct_supplierid_productid PRIMARY KEY
(SupplierID, ProductID),

    CONSTRAINT fk_supplierproduct_supplier_id FOREIGN KEY (SupplierID)
REFERENCES Supplier(SupplierID),

    CONSTRAINT fk_supplierproduct_product_id FOREIGN KEY (ProductID)
REFERENCES Product(ProductID)

);


CREATE TABLE SupplierOrder (

    OrderID             INT NOT NULL AUTO_INCREMENT COMMENT 'Unique
supplier order identifier',

    OrderDate           DATE NOT NULL COMMENT 'Date the order was created',
```

```
    ExpectedDeliveryDate DATE COMMENT 'Expected date at the destination warehouse',
    Status               VARCHAR(20) NOT NULL DEFAULT 'Draft' COMMENT 'Lifecycle status; Draft and Submitted are used by the report rules',
    SupplierID           INT NOT NULL COMMENT 'Supplier receiving the purchase order',
    WarehouseID          INT NOT NULL COMMENT 'Destination warehouse for the order',
    CONSTRAINT pk_order_id PRIMARY KEY (OrderID),
    CONSTRAINT fk_order_supllier_id FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID),
    CONSTRAINT fk_order_warehouse_id FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID)
);
CREATE INDEX idx_order_status_date ON SupplierOrder(Status, OrderDate) COMMENT 'Supports filtering orders by status and date';

CREATE TABLE SupplierOrderItem (
    OrderItemID INT NOT NULL AUTO_INCREMENT COMMENT 'Unique order line identifier',
    OrderID     INT NOT NULL COMMENT 'Supplier order containing this line',
    ProductID   INT NOT NULL COMMENT 'Product being ordered',
    Quantity    DECIMAL(12,2) NOT NULL CHECK (Quantity > 0) COMMENT 'Quantity requested on this order line',
    UnitPrice   DECIMAL(12,2) NOT NULL CHECK (UnitPrice >= 0) COMMENT 'Historical price agreed for this order line',

    CONSTRAINT pk_orderitem_id PRIMARY KEY (OrderItemID),
    CONSTRAINT uq_orderitem_orderid_productid UNIQUE (OrderID, ProductID),
    CONSTRAINT fk_orderitem_order_id FOREIGN KEY (OrderID) REFERENCES SupplierOrder(OrderID),
    CONSTRAINT fk_orderitem_product_id FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
```

```
);


CREATE TABLE Shipment (
    ShipmentID      INT NOT NULL AUTO_INCREMENT COMMENT 'Unique shipment identifier',
    OrderID         INT NOT NULL COMMENT 'Supplier order fulfilled by this shipment',
    TrackingNumber  VARCHAR(50) COMMENT 'External carrier tracking reference',
    ShipmentDate    DATE COMMENT 'Date the shipment is dispatched',
    EstimatedArrival DATETIME COMMENT 'Current estimated arrival date and time',
    ActualArrival   DATETIME COMMENT 'Confirmed arrival date and time',
    Status          VARCHAR(20) NOT NULL DEFAULT 'Draft' COMMENT 'Shipment lifecycle status',


CONSTRAINT pk_shipment_id PRIMARY KEY (ShipmentID),
    CONSTRAINT fk_shipment_order_id FOREIGN KEY (OrderID) REFERENCES SupplierOrder(OrderID)
);
CREATE INDEX idx_shipment_status ON Shipment(Status);


CREATE TABLE ShipmentItem (
    ShipmentItemID INT NOT NULL AUTO_INCREMENT COMMENT 'Unique shipment line identifier',
    ShipmentID     INT NOT NULL COMMENT 'Shipment containing this allocation',
    OrderItemID    INT NOT NULL COMMENT 'Supplier order line being delivered',
    Quantity       DECIMAL(12,2) NOT NULL CHECK (Quantity > 0) COMMENT 'Quantity allocated to this shipment',


    CONSTRAINT pk_shipmentitem_id PRIMARY KEY (ShipmentItemID),
    CONSTRAINT fk_shipmentitem_shipment_id FOREIGN KEY (ShipmentID) REFERENCES Shipment(ShipmentID),
```

```
    CONSTRAINT fk_shipmentitem_orderitem_id FOREIGN KEY (OrderItemID)
REFERENCES SupplierOrderItem(OrderItemID)

);


CREATE TABLE TrackingEvent (

TrackingEventID INT NOT NULL AUTO_INCREMENT COMMENT 'Unique identifier
for a tracking event, including events at the same time',

    ShipmentID      INT NOT NULL COMMENT 'Shipment associated with this event',

    EventTime       DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
COMMENT 'Event time',

    Location        VARCHAR(150) COMMENT 'Location where the event was
recorded',

    Status          VARCHAR(30) NOT NULL COMMENT 'Status reported by the
tracking event',

    Description     TEXT COMMENT 'Additional tracking or handling details',


    CONSTRAINT pk_trackingevent_id PRIMARY KEY (TrackingEventID),

    CONSTRAINT fk_trackingevent_shipment_id FOREIGN KEY (ShipmentID)
REFERENCES Shipment(ShipmentID)

);

CREATE INDEX idx_tracking_shipment_time ON TrackingEvent(ShipmentID,
EventTime, TrackingEventID) COMMENT 'Supports chronological tracking history
per shipment';


CREATE TABLE ProductComponent (

    ParentProductID     INT NOT NULL COMMENT 'Product assembled from
components',

    ComponentProductID INT NOT NULL COMMENT 'Direct component used by the
parent product',

    QuantityRequired    DECIMAL(12,2) NOT NULL CHECK (QuantityRequired > 0)
COMMENT 'Number of component units required for one parent unit',
```

```
    CONSTRAINT pk_productcomponent_id PRIMARY KEY (ParentProductID,
ComponentProductID),

    CONSTRAINT ck_productcomponent CHECK (ParentProductID <>
ComponentProductID),

    CONSTRAINT fk_productcomponent_parent_id FOREIGN KEY
(ParentProductID) REFERENCES Product(ProductID),

    CONSTRAINT fk_productcomponent_component_id FOREIGN KEY
(ComponentProductID) REFERENCES Product(ProductID)

);


CREATE TABLE StockMovement (

    MovementID      INT NOT NULL AUTO_INCREMENT COMMENT 'Unique stock
transaction identifier',

    InventoryID     INT NOT NULL COMMENT 'Inventory balance affected by this
movement',

    ShipmentItemID INT UNIQUE COMMENT 'Required only for Receipt movements;
each shipment item can be received once',

    MovementTime    DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
COMMENT 'Time the movement is recorded',

    MovementType    VARCHAR(20) NOT NULL COMMENT 'Opening, Receipt, Issue,
or Adjustment',

    QuantityDelta  DECIMAL(12,2) NOT NULL COMMENT 'Signed change in
inventory quantity',


    CONSTRAINT uq_stockmovementid_shipmentitemid UNIQUE(MovementID,
ShipmentItemID),

CONSTRAINT pk_stockmovement_id PRIMARY KEY(MovementID),

    CONSTRAINT fk_stockmovement_inventory_id FOREIGN KEY (InventoryID)
REFERENCES Inventory(InventoryID),

    CONSTRAINT fk_stockmovement_shipmentitem_id FOREIGN KEY
(ShipmentItemID) REFERENCES ShipmentItem(ShipmentItemID)

);
```

```
CREATE INDEX idx_movement_inventory_time ON StockMovement(InventoryID, MovementTime, MovementID) COMMENT 'Supports movement history lookup by inventory and time';


CREATE TABLE RestockAlert (

AlertID              INT NOT NULL AUTO_INCREMENT COMMENT 'Unique low-stock alert period identifier',

   InventoryID          INT NOT NULL COMMENT 'Inventory record that triggered the alert',

   AlertDate            DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP COMMENT 'Time the low-stock condition was detected',

   QuantityAtAlert      DECIMAL(12,2) NOT NULL COMMENT 'Quantity snapshot when this alert opened',

   ReorderLevelAtAlert  DECIMAL(12,2) NOT NULL COMMENT 'Reorder level snapshot when this alert opened',

   ResolvedAt           DATETIME COMMENT 'Resolution time; NULL means alert is unresolved',

   Message              VARCHAR(200) COMMENT 'Human-readable alert explanation',


   CONSTRAINT pk_restockalert_id PRIMARY KEY(AlertID),

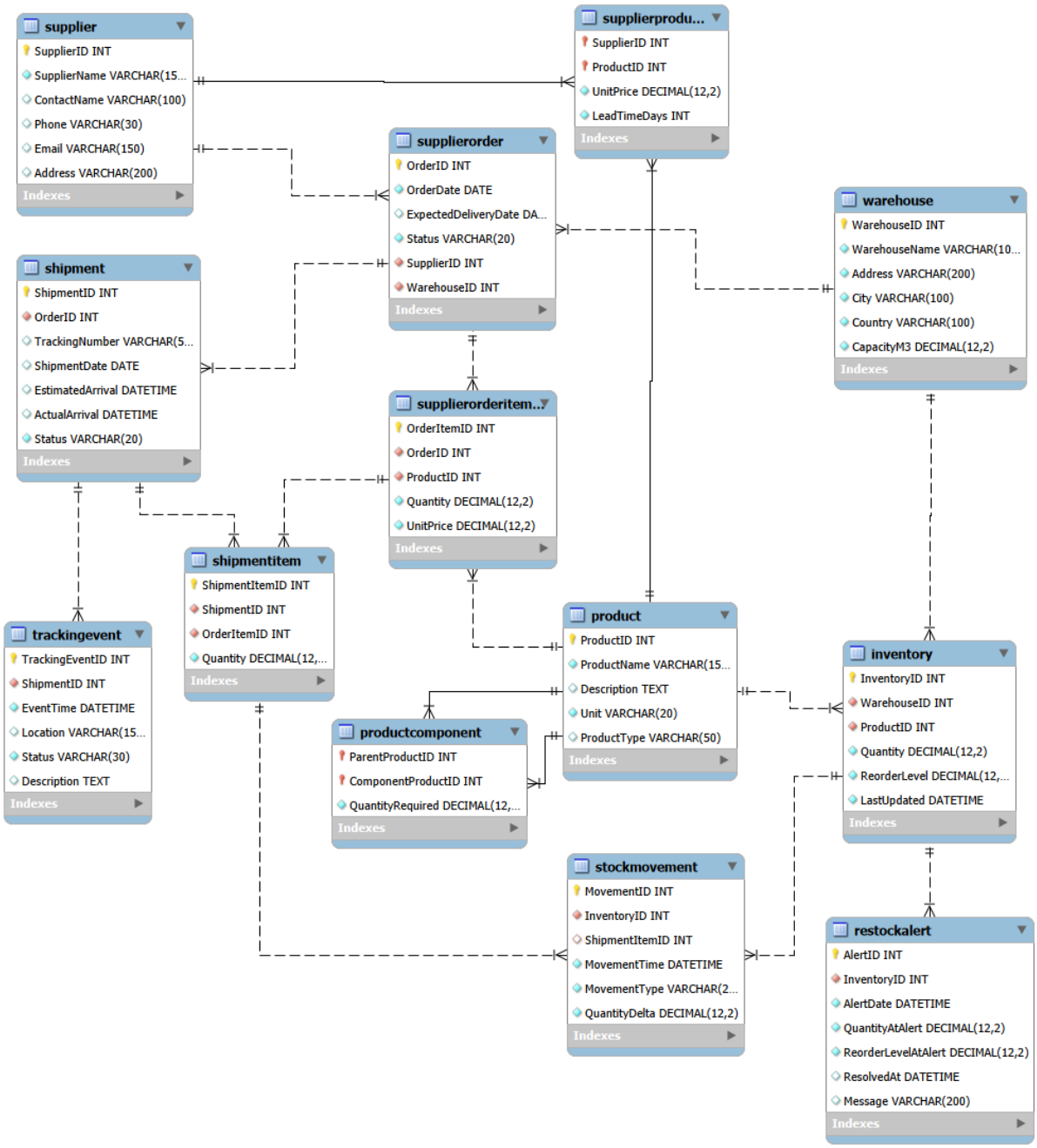
CONSTRAINT ck_restockalert CHECK (QuantityAtAlert < ReorderLevelAtAlert),

   CONSTRAINT fk_restockalert_inventory_id FOREIGN KEY (InventoryID) REFERENCES Inventory(InventoryID)

);

CREATE INDEX idx_alert_inventory_date ON RestockAlert(InventoryID, AlertDate) COMMENT 'Supports alert-history lookup for an inventory record';
```

II/ Entity-Relationship Diagrams (ERDs)



III/ Table Descriptions

Table 1. WAREHOUSE

1/ Purpose: Stores information about warehouses, including their names, locations, and storage capacities.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description

WarehouseID	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier of a warehouse.
WarehouseName	VARCHAR(100)	NOT NULL	Name of the warehouse.
Address	VARCHAR(200)	NOT NULL	Physical address of the warehouse.
City	VARCHAR(100)	NOT NULL	City where the warehouse is located.
Country	VARCHAR(100)	NOT NULL	Country where the warehouse is located.
CapacityM3	DECIMAL(12,2)	NOT NULL, CHECK (CapacityM3>=0)	Maximum storage capacity in cubic meters. Must not be negative.

Table 2. PRODUCT

1/ Purpose: Stores information about products that can be purchased, stored, or assembled within the supply chain.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
ProductID	INT	PK, NOT NULL	Unique identifier of a product.
ProductName	VARCHAR(150)	NOT NULL	Name used to identify the product.
Description	TEXT	NULL	Additional descriptive information about the product.
Unit	VARCHAR(20)	NOT NULL	Base unit used to measure the product quantity.
ProductType	VARCHAR(50)	NULL	Type or category of the product. Controlled list is finalized in Phase 2.

Table 3. INVENTORY

1/ Purpose: Records the current stock quantity and reorder level of each product at each warehouse.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
InventoryID	INT	PK, NOT NULL	Unique identifier of an inventory record.

WarehouseID	INT	FK, NOT NULL	References Warehouse. Each inventory belongs to exactly one warehouse.
ProductID	INT	FK, NOT NULL	References Product. Each inventory belongs to exactly one product.
Quantity	DECIMAL(12,2)	NOT NULL, CHECK (Quantity>=0)	Current quantity available. Cannot be negative.
ReorderLevel	DECIMAL(12,2)	NOT NULL, CHECK (ReorderLevel>=0)	Minimum inventory threshold. Cannot be negative.
LastUpdated	DATETIME	NOT NULL	Date and time when the inventory balance was most recently changed.
Additional constraint : UNIQUE (WarehouseID, ProductID)			

Table 4. SUPPLIER

1/ Purpose: Stores information about suppliers that provide products to the organization.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
SupplierID	INT	PK, NOT NULL	Unique identifier of a supplier.
SupplierName	VARCHAR(150)	NOT NULL	Name of the supplier.
ContactName	VARCHAR(100)	NULL	Name of the main contact person.
Phone	VARCHAR(30)	NULL	Supplier's contact telephone number.
Email	VARCHAR(150)	NULL	Supplier's contact email address.
Address	VARCHAR(200)	NULL	Supplier's business address.

Table 5. SUPPLIER_PRODUCT

1/ Purpose: Records the products offered by each supplier, including current unit prices and lead times.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description

SupplierID	INT	PK, FK, NOT NULL	References Supplier and forms part of the composite primary key.
ProductID	INT	PK, FK, NOT NULL	References Product and forms part of the composite primary key.
UnitPrice	DECIMAL(12,2)	NOT NULL, CHECK (UnitPrice >= 0)	Current price offered by the supplier.
LeadTimeDays	INT	NOT NULL, CHECK (LeadTimeDays >= 0)	Expected number of days required for delivery.

Table 6. SUPPLIER_ORDER

1/ Purpose: Stores purchase order information, including the supplier, destination warehouse, and order status.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
OrderID	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier of a supplier order.
OrderDate	DATE	NOT NULL	Date on which the supplier order is created.
ExpectedDeliveryDate	DATE	NULL	Expected delivery date at the destination warehouse.
Status	VARCHAR(20)	NOT NULL	Current lifecycle state of the supplier order.
SupplierID	INT	FK, NOT NULL	References Supplier. Each order refers to exactly one supplier.
WarehouseID	INT	FK, NOT NULL	References Warehouse. Each order has exactly one destination warehouse.

Table 7. SUPPLIER_ORDER_ITEM

1/ Purpose: Records individual product lines in a supplier order, including the ordered quantity and agreed unit price.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
OrderItemID	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier of an individual order line.
OrderID	INT	FK, NOT NULL	References SupplierOrder.
ProductID	INT	FK, NOT NULL	References Product.
Quantity	DECIMAL(12,2)	NOT NULL, CHECK (Quantity > 0)	Quantity of the product requested in the order line.
UnitPrice	DECIMAL(12,2)	NOT NULL, CHECK (UnitPrice >= 0)	Historical price agreed when the order was created.
Additional constraint : UNIQUE (OrderID, ProductID)			

Table 8. SHIPMENT

1/ Purpose: Stores information about shipments associated with supplier orders, including shipment dates, tracking numbers, and delivery status.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
ShipmentID	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier of a shipment.
OrderID	INT	FK, NOT NULL	References SupplierOrder. One order may have multiple shipments.
TrackingNumber	VARCHAR(50)	NULL	External tracking reference assigned to the shipment.
ShipmentDate	DATE	NULL	Date when the shipment is dispatched.
EstimatedArrival	DATETIME	NULL	Current expected arrival date and time.
ActualArrival	DATETIME	NULL	Confirmed arrival date and time. Empty until arrival is confirmed.
Status	VARCHAR(20)	NOT NULL	Current logistics state of the shipment.

Table 9. SHIPMENT_ITEM

1/ Purpose: Records the quantity of each supplier order item allocated to a shipment and supports partial deliveries.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
ShipmentItemID	INT	PK, NOT NULL	Unique identifier of a shipment line.
ShipmentID	INT	FK, NOT NULL	References Shipment.
OrderItemID	INT	FK, NOT NULL	References SupplierOrderItem. Must belong to the same supplier order as the shipment.
Quantity	DECIMAL(12,2)	NOT NULL, CHECK (Quantity > 0)	Quantity allocated to this shipment. Must be positive. Total allocation cannot exceed the originally ordered quantity.

Table 10. TRACKING_EVENT

1/ Purpose: Records shipment tracking events, including their timestamps, locations, statuses, and descriptions.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
TrackingEventID	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier of a tracking event.
ShipmentID	INT	FK, NOT NULL	References Shipment.
EventTime	DATETIME	NOT NULL	Date and time when the tracking event occurs.
Location	VARCHAR(150)	NULL	Geographical location where the event is recorded.
Status	VARCHAR(30)	NOT NULL	Shipment status reported by the tracking event.
Description	TEXT	NULL	Additional information about the event or handling activity.

Table 11. PRODUCT_COMPONENT

1/ Purpose: Defines the component products and required quantities used to make or assemble parent products, supporting multi-level Bills of Materials (BOM).

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
ParentProductID	INT	PK, FK, NOT NULL	References Product as the parent/assembled product.
ComponentProductID	INT	PK, FK, NOT NULL	References Product as the component product.
QuantityRequired	DECIMAL(12,2)	NOT NULL, CHECK (QuantityRequired > 0)	Number of component units required to produce one parent product. Must be greater than zero.
Additional constraint : CHECK (ParentProductID <> ComponentProductID)			

Table 12. STOCK_MOVEMENT

1/ Purpose: Records inventory transactions, such as receipts, issues, and adjustments, to maintain a history of stock changes.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
MovementID	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier of an inventory transaction.
InventoryID	INT	FK, NOT NULL	References Inventory.
ShipmentItemID	INT	FK, NULL	Optional reference to ShipmentItem. Used only for Receipt movements.
MovementTime	DATETIME	NOT NULL	Date and time when the inventory transaction is recorded.
MovementType	VARCHAR(20)	NOT NULL	Category of inventory change, e.g. Opening,

			Receipt, Issue, Adjustment.
QuantityDelta	DECIMAL(12,2)	NOT NULL	Signed inventory change. Positive increases inventory; negative decreases inventory.

Table 13. RESTOCK_ALERT

1/ Purpose: Records low-stock alert periods when inventory falls below its reorder level and tracks when those alerts are resolved.

2/ Table structure:

Attribute	Data Type	Constraints / Notes	Description
AlertID	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier of a low-stock alert period.
InventoryID	INT	FK, NOT NULL	References Inventory.
AlertDate	DATETIME	NOT NULL	Date and time when the low-stock condition is first detected.
QuantityAtAlert	DECIMAL(12,2)	NOT NULL	Inventory quantity captured when the alert is created. Historical snapshot.
ReorderLevelAtAlert	DECIMAL(12,2)	NOT NULL	Reorder level captured when the alert is created. Historical snapshot.
ResolvedAt	DATETIME	NULL	Date and time when the alert is resolved. NULL means unresolved.
Message	VARCHAR(200)	NULL	Human-readable explanation of the low-stock condition.

IV/ Index Documentation

1/ Index List

Index Name	Indexed Columns	Index Type	Purpose
idx_warehouse_name	WarehouseName	Non-unique	Supports searching warehouses by name.
idx_product_name	ProductName	Non-unique	Supports searching products by name.
idx_supplier_name	SupplierName	Non-unique	Supports searching suppliers by name.
idx_order_status_date	Status, OrderDate	Composite, non-unique	Supports filtering supplier orders by status and date.
idx_shipment_status	Status	Non-unique	Supports filtering shipments by status.
idx_tracking_shipment_time	ShipmentID, EventTime, TrackingEventID	Composite, non-unique	Supports retrieving tracking events in chronological order for a shipment.
idx_movement_inventory_time	InventoryID, MovementTime, MovementID	Composite, non-unique	Supports retrieving inventory movement history by inventory record and time.
idx_alert_inventory_date	InventoryID, AlertDate	Composite, non-unique	

V/ Constraints Documentation

1/ Primary Key Constraints

Table	Primary Key	Purpose
Warehouse	`WarehouseID`	Uniquely identifies each warehouse.
Product	`ProductID`	Uniquely identifies each product.
Supplier	`SupplierID`	Uniquely identifies each supplier.
Inventory	`InventoryID`	Uniquely identifies each inventory record.
SupplierProduct	`SupplierID`, `ProductID`	Identifies each supplier-product association.
SupplierOrder	`OrderID`	Uniquely identifies each supplier order.
SupplierOrderItem	`OrderItemID`	Uniquely identifies each supplier order line.
Shipment	`ShipmentID`	Uniquely identifies each shipment.
ShipmentItem	`ShipmentItemID`	Uniquely identifies each shipment line.
TrackingEvent	`TrackingEventID`	Uniquely identifies each tracking event.
ProductComponent	`ParentProductID`, `ComponentProductID`	Identifies each parent-product and component-product relationship.
StockMovement	`MovementID`	Uniquely identifies each stock movement.
RestockAlert	`AlertID`	Uniquely identifies each restock alert period.

2/ Foreign Key Constraints

Constraint Name	Table	Foreign Key Column	Referenced Table and Column
`fk_inventory_warehouse_id`	Inventory	`WarehouseID`	Warehouse(`WarehouseID`)
`fk_inventory_product_id`	Inventory	`ProductID`	Product(`ProductID`)
`fk_supplierproduct_supplier_id`	SupplierProduct	`SupplierID`	Supplier(`SupplierID`)

fk_supplierproduct_product_id	SupplierProduct	ProductID	Product(ProductID)
fk_order_supplier_id	SupplierOrder	SupplierID	Supplier(SupplierID)
fk_order_warehouse_id	SupplierOrder	WarehouseID	Warehouse(WarehouseID)
fk_orderitem_order_id	SupplierOrderItem	OrderID	SupplierOrder(OrderID)
fk_orderitem_product_id	SupplierOrderItem	ProductID	Product(ProductID)
fk_shipment_order_id	Shipment	OrderID	SupplierOrder(OrderID)
fk_shipmentitem_shipment_id	ShipmentItem	ShipmentID	Shipment(ShipmentID)
fk_shipmentitem_orderitem_id	ShipmentItem	OrderItemID	SupplierOrderItem(OrderItemID)
fk_trackingevent_shipment_id	TrackingEvent	ShipmentID	Shipment(ShipmentID)
fk_productcomponent_parent_id	ProductComponent	ParentProductID	Product(ProductID)
fk_productcomponent_component_id	ProductComponent	ComponentProductID	Product(ProductID)
fk_stockmovement_inventory_id	StockMovement	InventoryID	Inventory(InventoryID)
fk_stockmovement_shipmentitem_id	StockMovement	ShipmentItemID	ShipmentItem(ShipmentItemID)
fk_restockalert_inventory_id	RestockAlert	InventoryID	

3/ Unique Constraints

Constraint Name	Table	Columns	Purpose
uq_inventory_warehouseid_productid	Inventory	WarehouseID, ProductID	Ensures that each product has at most one inventory record in a given

			warehouse.
`uq_orderitem_orderid_productid`	SupplierOrderItem	`OrderID, ProductID`	Prevents the same product from appearing in multiple order lines within the same supplier order.
Unique constraint on `ShipmentItemID`	StockMovement	`ShipmentItemID`	Allows a shipment item to be associated with at most one stock movement through this column.

4/ Check Constraints

Table	Column(s)	Check Condition	Purpose
Warehouse	`CapacityM3`	`CapacityM3 >= 0`	Prevents negative warehouse capacity.
Inventory	`Quantity`	`Quantity >= 0`	Prevents negative inventory balances.
Inventory	`ReorderLevel`	`ReorderLevel >= 0`	Prevents negative reorder thresholds.

SupplierProduct	`UnitPrice`	`UnitPrice >= 0`	Prevents negative supplier prices.
SupplierProduct	`LeadTimeDays`	`LeadTimeDays >= 0`	Prevents negative delivery lead times.
SupplierOrderItem	`Quantity`	`Quantity > 0`	Ensures that the ordered quantity is positive.
SupplierOrderItem	`UnitPrice`	`UnitPrice >= 0`	Prevents negative order-line prices.
ShipmentItem	`Quantity`	`Quantity > 0`	Ensures that the allocated shipment quantity is positive.
ProductComponent	`QuantityRequired`	`QuantityRequired > 0`	Ensures that the required component quantity is positive.
ProductComponent	`ParentProductID, ComponentProductID`	`ParentProductID <> ComponentProductID`	Prevents a product from being its own direct component.
RestockAlert	`QuantityAtAlert, ReorderLevelAtAlert`	`QuantityAtAlert < ReorderLevelAtAlert`	Ensures that an alert represents a stock level below

			the recorded reorder threshold.

VI/ Stored Procedures and Functions Documentation

1/ Stored Procedure: sp_SubmitSupplierOrder

Purpose: Submits a supplier order.
Parameters:
- p_OrderID: The ID of the supplier order to submit.
Description: This procedure checks whether the supplier order exists and contains at least one order item. If the conditions are satisfied, the order status is updated to Submitted.
SQL Code:

```
DELIMITER //
CREATE PROCEDURE sp_SubmitSupplierOrder(IN p_OrderID INT)
BEGIN
   UPDATE SupplierOrder
   SET Status = 'Submitted'
   WHERE OrderID = p_OrderID;
END //

DELIMITER;
```

2/ Stored Procedure: sp_AddShipmentItem

Purpose: Adds a product item to a shipment.
Parameters:
-p_ShipmentID: The ID of the shipment.
-p_OrderItemID: The ID of the supplier order item.
-p_Quantity: The quantity of the item.
Description: This procedure inserts a new record into ShipmentItem, linking an order item to a shipment.
SQL Code:

```
DELIMITER //
CREATE PROCEDURE sp_AddShipmentItem(IN p_ShipmentID INT, IN
p_OrderItemID INT, IN p_Quantity INT)
BEGIN
   INSERT INTO ShipmentItem(ShipmentID, OrderItemID, Quantity)
   VALUES (p_ShipmentID, p_OrderItemID, p_Quantity);
```

```
END //

DELIMITER ;
```

3/ Stored Procedure: sp_RecordStockMovement

Purpose: Records a stock movement.
Parameters:
-p_InventoryID: The ID of the inventory record.
-p_ShipmentItemID: The related shipment item ID; can be NULL.
-p_MovementType: The type of stock movement.
-p_QuantityDelta: The change in inventory quantity.
Description: This procedure inserts a record into StockMovement to maintain the history of inventory changes.
SQL Code:

```
DELIMITER //
CREATE PROCEDURE sp_RecordStockMovement(IN p_InventoryID INT, IN
p_ShipmentItemID INT, IN p_MovementType VARCHAR(20), IN p_QuantityDelta
INT)
BEGIN
   INSERT INTO StockMovement(InventoryID, ShipmentItemID, MovementTime,
MovementType, QuantityDelta)
   VALUES(p_InventoryID, p_ShipmentItemID, CURRENT_TIMESTAMP,
p_MovementType, p_QuantityDelta);
END //

DELIMITER ;
```

4/ Stored Procedure: sp_ReceiveShipmentItem

Purpose: Records the receipt of a shipment item.
Parameters:
-p_ShipmentItemID: The ID of the received shipment item.
-p_InventoryID: The ID of the inventory record to update.
Description: This procedure retrieves the quantity of a shipment item and records it as an inbound stock movement.
SQL Code:

```
DELIMITER //
CREATE PROCEDURE sp_ReceiveShipmentItem(IN p_ShipmentItemID INT, IN
p_InventoryID INT)
BEGIN
   DECLARE v_Quantity INT;
```

```
    SELECT Quantity INTO v_Quantity
    FROM ShipmentItem
    WHERE ShipmentItemID = p_ShipmentItemID;
    UPDATE Inventory
    SET Quantity = Quantity + v_Quantity,
        LastUpdated = CURRENT_TIMESTAMP
    WHERE InventoryID = p_InventoryID;
    INSERT INTO StockMovement(InventoryID, ShipmentItemID, MovementTime,
MovementType, QuantityDelt)
    VALUES(p_InventoryID, p_ShipmentItemID,CURRENT_TIMESTAMP, 'IN',
v_Quantity);
END //

DELIMITER ;
```

5/ Stored Procedure: sp_SetInventoryReorderLevel

Purpose: Updates the reorder level of an inventory record.
Parameters:
-p_InventoryID: The ID of the inventory record.
-p_ReorderLevel: The new reorder level.
Description: This procedure updates the minimum stock threshold in the Inventory table.
SQL Code:

```
DELIMITER //
CREATE PROCEDURE sp_SetInventoryReorderLevel( IN p_InventoryID INT, IN
p_ReorderLevel INT)
BEGIN
    UPDATE Inventory
    SET ReorderLevel = p_ReorderLevel
    WHERE InventoryID = p_InventoryID;
END //

DELIMITER ;
```

6/ Function: fn_GetInventoryQuantity

Purpose: Retrieves the current inventory quantity.
Parameters:
- p_InventoryID: The ID of the inventory record.
Description: This function returns the quantity of the specified inventory record.
SQL Code:

```
DELIMITER //
```

```
CREATE FUNCTION fn_GetInventoryQuantity(p_InventoryID INT)
RETURNS INT
READS SQL DATA
BEGIN
   DECLARE v_Quantity INT;
   SELECT Quantity INTO v_Quantity
   FROM Inventory
   WHERE InventoryID = p_InventoryID;
   RETURN v_Quantity;
END //

DELIMITER ;
```

7/ Function: fn_IsInventoryLow

Purpose: Checks whether inventory is below its reorder level.
Parameters:
- p_InventoryID: The ID of the inventory record.

Description: This function returns 1 if the inventory quantity is below the reorder level, and 0 otherwise.
SQL Code:

```
DELIMITER //
CREATE FUNCTION fn_IsInventoryLow(p_InventoryID INT)
RETURNS TINYINT
READS SQL DATA
BEGIN
   DECLARE v_Result TINYINT;
   SELECT Quantity < ReorderLevel INTO v_Result
   FROM Inventory
   WHERE InventoryID = p_InventoryID;
   RETURN v_Result;
END //

DELIMITER ;
```